Hitha Shri Nagaruru

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F-1 student at UCI (MCS, Dec 2026). CPT-eligible for internship roles. No sponsorship required for internship.

EDUCATION

- UNIVERSITY OF CALIFORNIA, IRVINE, Irvine, California
  Master of Computer Science, September 2025 - December 2026, GPA: 3.83/4.00
- JSS ACADEMY OF TECHNICAL EDUCATION, Bengaluru, India
  Bachelor of Engineering in Computer Science and Engineering, December 2020 - June 2024, GPA: 3.81/4.00

RECENT WORK EXPERIENCE

BOSCH LIMITED, Bengaluru, India
AI Engineer - Graduate Apprentice, *Digitization and Industry 4.0 department* *August 2024 - August 2025*

- Executed a full EDA pipeline on diesel pump calibration bench data, improving **productivity by 2.6%** and reducing **cycle time by 9 seconds**, avoiding **₹40M in capital expenditure** on new test benches.
- Developed a **GPT-4o–powered chatbot** for **real-time MES breakdown resolution** in the oxygen sensor line using system-generated error codes (no manual input), delivering **96% accuracy** on known failure cases with a **retrieval-based pipeline** and **100% safe fallback** for unseen errors, **reducing production downtime**.
- Built a **Daily Leadership Review chatbot** using Azure OpenAI (GPT-4o) and FastAPI, integrating directly with a **live Manufacturing Execution System** (Oracle) database to provide real-time KPI insights, **eliminating** the need for **150+ Power BI licenses**.
- Designed an end-to-end **automated data pipeline** converting **machine-generated PDFs** from shared network storage into structured Excel datasets using OCR (Tesseract, PyMuPDF) and Pandas, integrated with **Power BI** to enable **real-time OK/NOK tracking** and faster defective part quarantine.
- **Deployed the first Python-based on-premise OCR web application** within the plant network using Flask (served via Waitress/NSSM), enabling **secure** digitization of **sensitive HR documents without external data exposure**.

TEQUED LABS, Bengaluru, India
AI & ML intern *August 2023 - September 2023*

- Analyzed the **Tokyo 2021 Olympics dataset** using Python (NumPy, Pandas, Matplotlib), uncovering **trends** in medal distribution, gender representation, host nation advantage, participation in newly introduced sports (e.g., skateboarding, surfing), and the **impact of COVID-19**, delivering structured insights through **exploratory data analysis**.

VOIS - VODAFONE INTELLIGENT SOLUTIONS, Remote
Intern *August 2022 - October 2022*

- Developed a **real-time helmet detection system** using YOLOv3 and OpenCV to identify non-compliant riders from CCTV footage, enabling **automated traffic violation detection**.
- Implemented vehicle number plate logging for non-compliant cases and **automated reporting** via Excel, streamlining **enforcement workflows** and **reducing manual monitoring effort**.

PROJECTS

VALIDATION OF LLMS | Documentation | Published package **January 2026 - Present**

- Led a 4-person team to architect and publish an **open-source Python package** (**validate-llm**) to PyPI, delivering five composable **guardrail agents** (toxicity, privacy, accuracy, relevancy, bias) that wrap **any LLM pipeline** with structured PASS/FAIL verdicts and calibrated 0-1 scoring, supporting multiple providers via **LiteLLM**.
- Engineered a **3-layer toxicity detection system** combining **rule-based profanity filtering**, **Detoxify ML scoring**, and **SentenceTransformer** (all-MiniLM-L6-v2) semantic similarity classification across policy categories (hate speech, violence, extremism, self-harm), achieving **96% detection accuracy** with fully local inference and **zero external API calls.**
- Designed the framework with a **plug-and-play RAG interface**, enabling developers in production settings to integrate their own vector store retriever for **domain-specific factual grounding**; built the accuracy agent using LLM-as-a-judge via **DeepEval GEval** with **BM25-ranked web retrieval** as a default evidence source, reaching **85% accuracy** on benchmarks, and shipped a full-stack demo (FastAPI + SSE streaming + web UI) with a **hosted documentation site**.

FEDERATED LEARNING ENGINE WITH PRIVACY RISK DETECTION AND PROTECTION **March 2026 - Present**

- Designing a **federated learning system** simulating decentralized clients with **non-IID data distributions**, implementing **FedAvg**-based model aggregation.
- Building a **membership inference attack** that uses model prediction confidence to identify potential training data leakage.
- Integrating **differential privacy mechanisms (gradient clipping, noise injection)** directly into the training pipeline to mitigate leakage.
- Measuring **privacy–utility tradeoffs** by analyzing attack success rates against model accuracy under varying noise levels.

SKILLS

- **Generative AI & NLP:** Azure OpenAI, Anthropic Claude, RAG, Prompt Engineering, FastAPI, LLM Evaluation, Anthropic Claude.
- **ML & AI:** Python, PyTorch, TensorFlow, Scikit-learn, CUDA, Hugging Face, Anomaly Detection, Hypothesis Testing.
- **Data & Engineering:** SQL (PostgreSQL, Oracle, MySQL), PySpark, Databricks, Azure ML Studio, Power BI, ETL/ELT, Git, Tableau.